GLOBAL SCHOOL OF SCIENCE

First Terminal Examination - 2079

Subject: Chemistry (3011)

Class : XI | **Full Marks: 75**

Time: 3 hrs. | **SET 'A'** | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternatives. **[1 × 11 = 11]**

1. Which of the following configurations represent the element that shows flame colouration?

   a. $1s^2\ 2s^2\ 2p^1$ b. $1s^2\ 2s^2 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$

   c. $1s^2\ 2s^2 2p^6\ 3s^1$ d. $1s^2\ 2s^2 2p^6$

2. Which of the following is the correct order of atomic size for group 2 elements?

   a. Be > Mg > Ca > Sr b. Sr > Ca > Mg > Be

   c. Mg > Ca > Be > Sr d. Ca > Mg > Sr > Be

3. The smallest ion among the following is,

   a. Cl^- b. K^+ c. Ca^{++} d. S^{--}

4. Which of the following is not true about Mendeleev's periodic table?

   a. It is based on atomic mass

   b. It classified isotopes

   c. It corrected atomic masses of some elements

   d. It left gaps for undiscovered elements

5. Concept of quantization of angular momentum was introduced by

   a. Bohr b. Sommerfeld c. Dalton d. Heisenberg

6. Paschen series is obtained when electron of excited H – atom fall from higher orbits to …

a. first orbit
b. second orbit
c. third orbit
d. fourth orbit

7. Which of the following has negligible mass

a. nucleus
b. proton
c. neutron
d. electron

8. Which of the following is least metallic

a. P
b. As
c. Sb
d. Bi

9. Which of the following will not displace hydrogen from dilute HCl?

a. Ba
b. Pb
c. Hg
d. Sn

10. The adsorption of hydrogen by Pd is

a. Reduction
b. Occlusion
c. Hydrogenation
d. Dehydrogenation

11. The most abundant element on earth's surface is

a. Oxygen
b. Silicon
c. Aluminium
d. Iron

Group - B

Give short answer to the following questions. **[8 × 5 = 40]**

1. a. Differentiate between nuclear charge and effective nuclear charge. [1]

   b. Define atomic radius. What are the difficulties to find its absolute value? What is covalent radius? [1 +2 + 1]

2. a. The species Na^+, Mg^{++} and Al^{+++} ions are isoelectronic species. Compare their size with reasons. Why is Cl^- larger than Cl-atom? [2+1]

   b. Which one is larger in size K or K^+ and why? [2]

3. Rutherford cannot explain formation of line spectra in hydrogen and hydrogen like system.

   a. How does Bohr's theory predict the origin of line spectra of hydrogen atom? [2]

   b. Why do electrons not fall into nucleus according to Bohr's theory? [1]

   c. List out the limitations of Bohr's atomic theory. [2]

4. Bohr's atomic theory successfully explain the formation of spectra in hydrogen.

   a. Draw a figure of hydrogen spectrum to show spectral series and their region. [3]

   b. Explain the spectral line observed in visible region? [1]

   c. What is the meaning of "Stationary states" in Bohr's atomic model? [1]

5. a. What do you mean by the dual nature of an electron? Write expression to relate wave nature of electron with particle nature. Is the equation applied for macroparticle like bullet? [1+2]

   b. Why does hydrogen gas show large number of spectral series though hydrogen atom contains one electron? [2]

6. Oxygen is the first element of group 16 (VIA) in the periodic table. Its binary compounds are called oxides.

   a. What are amphoteric oxides? Give examples with suitable reason. [2]

   b. Why OF_2 is not considered as oxide? [1]

   c. Define allotropy. What are the allotropes of oxygen? [2]

7. Hydrogen and oxygen both are non-metals and are placed in separate groups in the periodic table.

   a. Why the position of hydrogen in the periodic table is controversial? [2]

   b. Oxygen group elements are regarded as chalcogens. Why? [1]

   c. How is atomic hydrogen prepared? [2]

8. The applications of oxides are associated with their properties. The key properties are mechanical, electrical and magnetic, thermal, biomedical etc. Some of the oxides are given below,

Fe_3O_4	NO	P_2O_5

   a. Classify the oxides from the above table on the basis of chemical behavior with the help of chemical reaction. [3]

   b. Mention the name and formula of oxides (anhydrides) of following acids: H_2SO_4, H_3PO_4, HVO_3 and HNO_3 [2]

Group - C

Give long answer to the following questions. **[3 × 8 = 24]**

9. a. What is a differentiating electron? Write characteristics of s- and p-block elements. [1+2]

   b. State modern periodic law. Draw an outline of the modern periodic table and locate i) representative elements ii) transition elements iii) inner transition elements and iv) noble gasses What are the demerits of modern periodic table? [1+2 + 2]

10. Different scientist gave different model to explain the structure of atom, which provides very useful information regarding the structure of atom.

    a. What is alpha particle? Write its charge and mass. [1]

    b. What are the observations of Rutherford's alpha particles scattering experiment? Write down the conclusions drawn from it about the structure of atom. [2+2]

    c. Which atomic model was improved by Rutherford's atomic model? [1]

    d. What is quantization of angular momentum? [2]

11. Hydrogen is the most abundant element in the universe. Different forms of hydrogen are known.

    a. Differentiate between nascent and atomic hydrogen. [2]

    b. What happens when a piece of zinc is dropped in aqueous solution of;

       i. Potassium permanganate acidified with dil. H_2SO_4?

       ii. Ferric chloride solution acidified with dil. HCl? [1.5+1.5]

    c. List out the isotopes of hydrogen with their number of nucleons. Also mention one use of each. [3]

- The End -